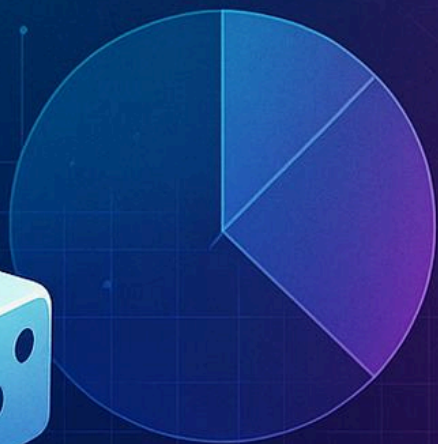


USEFUL FORMULAE & CONCEPT

MODERN MATHS

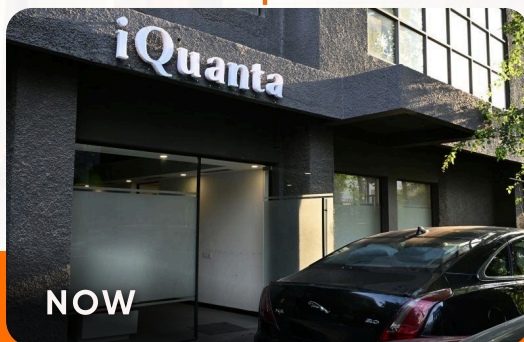
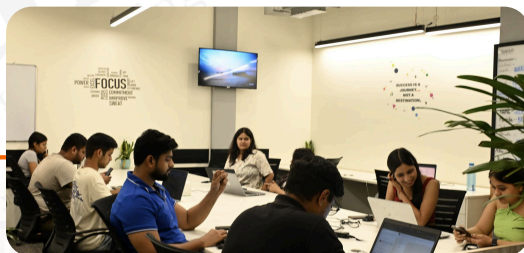
CAT QUANTITATIVE APTITUDE



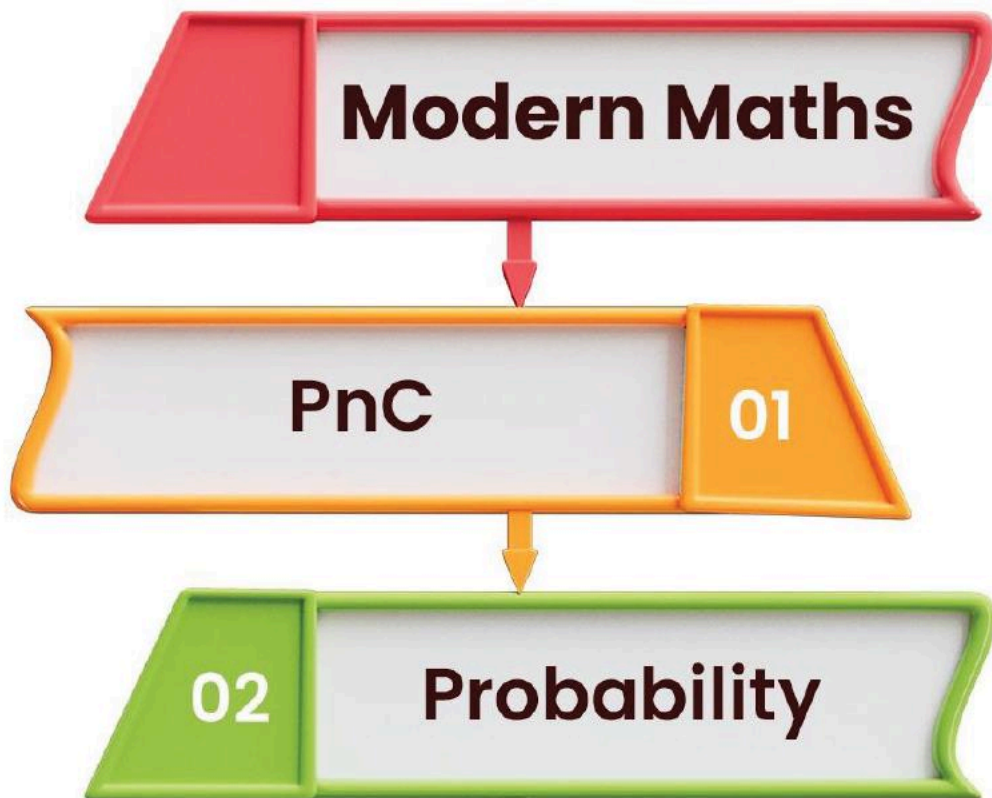
About iQuanta?

Founded in 2015 by Indrajeet Singh, iQuanta began as a Facebook doubt-solving group from a small apartment. With strong expertise in Quants, he earned the title 'Wizard of Quant' by solving thousands of students' queries with clarity and speed. Noticing the lack of quality guidance for CAT aspirants, he officially launched iQuanta to bridge this gap.

Over the years, iQuanta has continued to support over 4 lakh students through its active online communities, offering resources, and peer-driven learning. In the last 5 years, more than 8,000 students have converted IIMs. In 2024, iQuanta produced 500+ 99 percentilers, many of whom made it to top IIMs. iQuanta continues to guide thousands of aspirants on their journey to top B-schools.



Indrajeet Singh
Founder & CEO, iQuanta
Forbes 30 Under 30



PERMUTATION & COMBINATION



P&C Basics:

When two tasks are performed in succession, i.e., they are connected by an 'AND', to find the total number of ways of performing the two tasks, you must **MULTIPLY** the individual number of ways.

When only one of the two tasks is performed, i.e., the tasks are connected by an 'OR', to find the total number of ways of performing the two tasks you must **ADD** the individual number of ways.

• **Arrangement:** n items can be arranged in $n!$ ways

• **Permutation:** A way of selecting and arranging r objects out of a set of n objects. Denoted as ${}^n P_r = \frac{n!}{(n-r)!}$

• **Combination:** A way of selecting r objects out of n (arrangement does not matter) ${}^n C_r = \frac{n!}{(n-r)!r!}$

• Number of ways of selecting r things out of n distinct things is ${}^n C_r = {}^n C_{n-r}$

• No. of ways of choosing any number of things out of n distinct things is 2^n

• No. of ways of choosing any number of things out of n identical things is $n+1$

• No. of ways of distributing n identical things among r distinct groups such that all may get any number of things is ${}^{(n+r-1)}C_{(r-1)}$

IIM Ahmedabad & Bangalore Convert Rupesh Gupta shares his Journey with iQuanta!!

Rupesh Gupta
1 hr · 🌐

iQuanta
Building Learning Networks

"To be in the top 1%, you need to do what the other 99% won't!" - Anonymous

My Journey with CAT (and iQuanta) began in Sept 2019 when I left my job at Edelweiss to give my best attempt at CAT 2019. Even after putting my best foot forward, I screwed up on the exam day and managed to score 91%. The best call I could manage was SP Jain (Profile Based). I was convinced I wanted to re-attempt and pursue an MBA from the Holy Trinity.

Jump to CAT 2020, after 3 super-tedious months of prep, with constant support from **Jeet Singh** sir, **Akash Pandey** sir, and the entire iQuanta team, I managed to substantially improve my score and receive calls from major B-schools. The mentors work harder than you as the exam day approaches but they never get exhausted (unlike you). Due to iQuanta, I managed to score 99% in QA (My weakest section).

I was hysteric about getting calls from IIM - A/B. The days of 14-16 study hours seem like a distant memory now. My biggest takeaway is - Not focusing on Mock scores and only using them as stepping stones to master your preparation. I cannot underline enough the importance of self-belief in this tumultuous path. But you'll be glad when everything pays off in the end!!

#IIMA #IIMB #Converted

www.iQuanta.in

- No. of ways of distributing n identical things among r distinct groups such that all get at least 1 is ${}^{(n-1)}C_{(r-1)}$
 - No. of ways of distributing n distinct things among r distinct groups is r^n
 - If x items out of n items are repeated, then the number of ways of arranging these n items is $n!/x!$ ways. If a items, b items and c items are repeated within n items, they can be arranged in $\left(\frac{n!}{a!b!c!}\right)$ ways.
 - **Derangement:** If n distinct items are arranged, the number of ways they can be arranged so that they do not occupy their intended spot is

$$D = n! \left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots \dots \frac{(-1)^n}{n!} \right)$$
 - **Circular arrangement of 'n' distinct items:** Fix the first item and then arrange all the other items linearly with respect to the first item. This can be done in $(n-1)!$ ways.
- Note:** In a necklace, it can be done in $\frac{(n-1)!}{2}$ ways.

PROBABILITY



Probability is nothing but determining the chance, that event might occur. It is denoted by $P(E)$, where P is probability and E is the event.

$0 \leq P(E) \leq 1$ by definition

$$\text{Probability of event occurring} = \frac{\text{Number of favourable outcomes}}{\text{Total no. of outcomes}}$$

E.g. The **sample space** for the tossing of three coins simultaneously is given by:

$S = \{(T, T, T), (T, T, H), (T, H, T), (T, H, H), (H, T, T), (H, T, H), (H, H, T), (H, H, H)\}$

Total outcomes = $2 \times 2 \times 2 = 8$ since each coin can have 2 outcomes.

700+ 99%ilers & 1000s of CAT Toppers



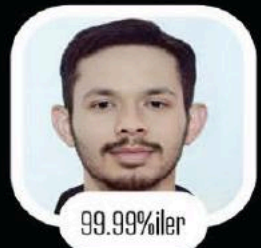
Rishi Mittal



Sabyasachi



Mridul Agarwal



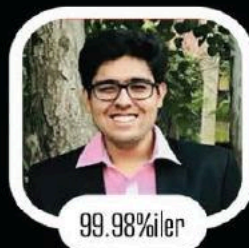
Parikshit Tomar



Guru Charan



Manav Jain



Shikhar Sachdeva



Mihir Kapse



Satyajit Das



Swadesh Rath



Kislay Jha



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Topper's Interview

iQuanta CAT 2024 IIM Converts

IIM Ahmedabad Converts



Vedant Chandewar
99.98%iler



Swaraj Pal
99.95%iler



Raunak Das
99.55%iler



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Mridul Tiwari
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Aman Ehtesham
99.94%iler

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99.98%iler



Sanvie Singhal
99.92%iler



Chitturi Sakshvith
Sanjeev
99.86%iler



Madhumitha
Sriharsha
99.6%iler

And many more

iQuanta vs OTHER COACHINGS

30,000 IIM Calls from iQuanta
CAT Course

COURSE FEATURES	CAT FULL COURSE	CAT FULL COURSE PRO	OTHER COACHINGS
Live Conceptual Classes	✓	✓	✓
Application Classes	✓	✓	✗
All Year Practice Sessions	✓	✓	✗
Assignments	✓	✓	✓
24*7 Doubt Solving	✓	✓	✗
Full Mocks with Video Solution	✓	✓	✓
Sectionals with Video Solution *	✓	✓	✗
Non-Engineer's Quant *	✓	✓	✗
Engineer's VARC *	✓	✓	✗
LRDI Inception *	✓	✓	✗
Ai Mock Analysis	✓	✓	✗
Special Initiatives * (QA250, LRDI 70)	✓	✓	✗
Peer to Peer Learning	✓	✓	✗
Free 2 Month Crash Course	✓	✓	✗
7500 CAT Qs in IIM ABC Course	✗	✓	✗
IIM ABC Practice Sessions (Daily 12 QA, 3RCs & 3 LRDI Sets)	✗	✓	✗
12 Sets: CAT Level Books *	✗	✓	✗

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99.5+%iler

